

## EXP 24:

Perform Sharpening of Image using High-Boost Masks.

### High-boost Masks

|    |         |    |    |         |    |
|----|---------|----|----|---------|----|
| 0  | -1      | 0  | -1 | -1      | -1 |
| -1 | $A + 4$ | -1 | -1 | $A + 8$ | -1 |
| 0  | -1      | 0  | -1 | -1      | -1 |

- $A \geq 1$
- if  $A = 1$ , it becomes "standard" Laplacian sharpening

**AIM:** Sharpening of Image using High-Boost Masks.

```
import cv2

import numpy as np

img = cv2.imread(r'C:\Users\pavan\OneDrive\Opencv\x.jpg')

if img is None:

    print("Error: Image not found or path is incorrect")

else:

    # Step 1: Apply Gaussian blur to reduce noise

    blurred = cv2.GaussianBlur(img, (3, 3), 0)

    # Step 2: Apply Laplacian operator

    laplacian = cv2.Laplacian(blurred, cv2.CV_64F)

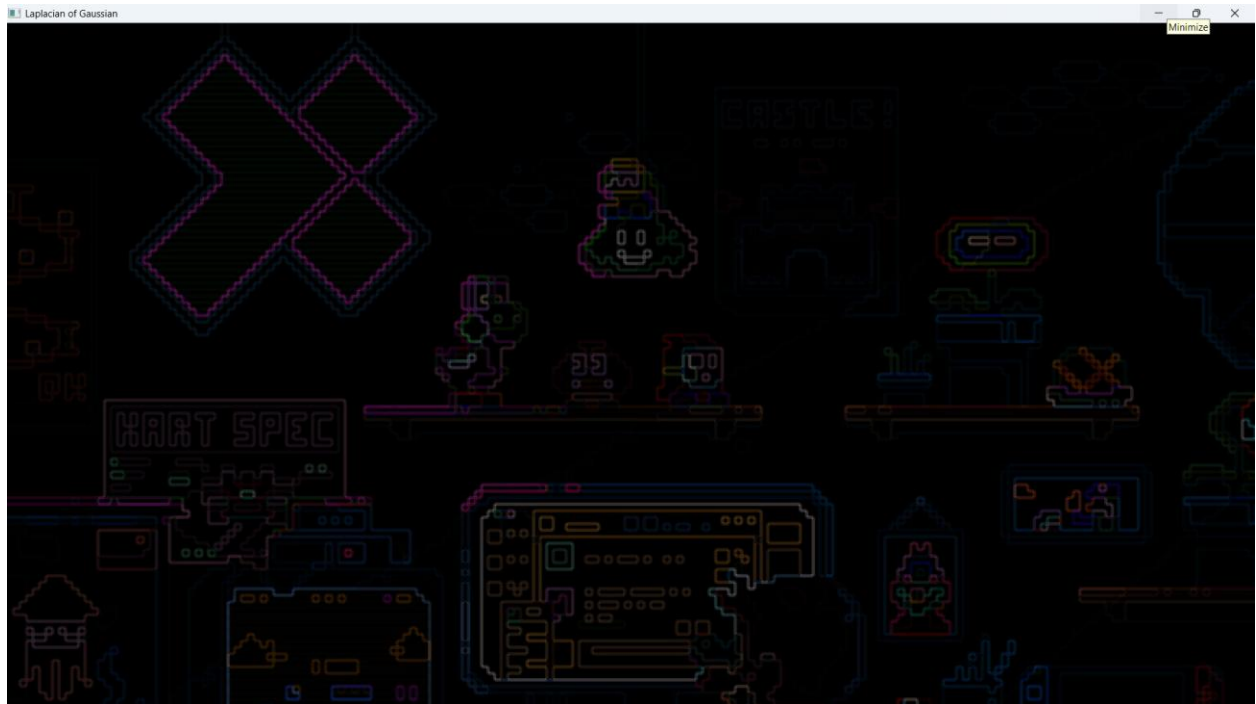
    # Convert back to uint8

    laplacian = cv2.convertScaleAbs(laplacian)
```

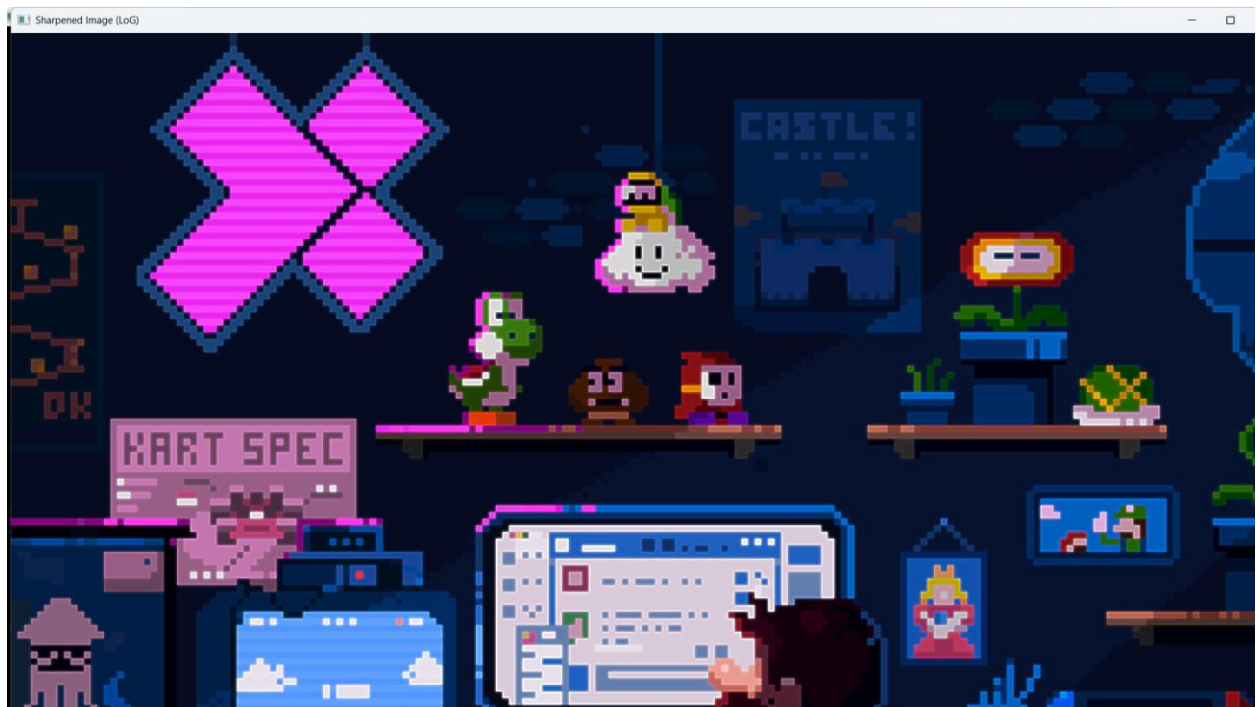
cv2.destroyAllWindows()

OUTPUT:

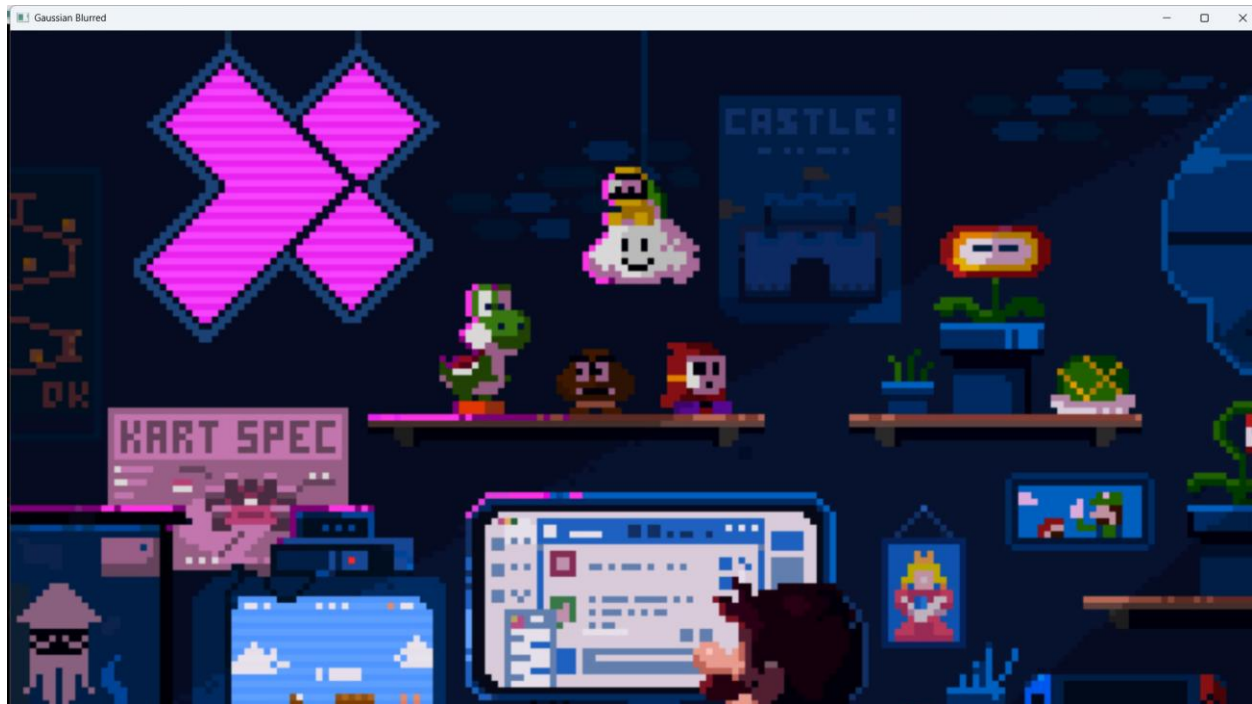
1)



2)



3)



Result:

Thus the program is executed successfully.